MILITARY SPECIFICATION

LEAK DETECTOR, FUEL SYSTEM

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers an explosion-proof leak detector for locating fuel leaks in aircraft fuel cells and tanks using helium, argon, or nitrous oxide as a tracer media.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-C-596	Connector, Electrical; Connector.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-636	Box, Shipping, Fiberboard.

MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-E-4158	Electronic Equipment Ground; General Requirements for.
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

*(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70 National Electrical Code

(Copies of the National Electrical Code may be obtained from the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Blvd, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 4.6).

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3.2 Components. The leak detector shall consist of the following major components.

Item	Quantity	Item name
1	1 each	Sensing unit
2	1 each	Power supply
3	1 each	Probe
4	1 each	Carrying case

3.3 General specification. The requirements of MIL-E-4158 apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4.1 Standard parts. AN or MS standard parts shall be used where they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, hose fittings, et cetera, may be used provided they have suitable properties and are interchangeable with AN, MS or MIL Standard parts without alterations.

3.4.2 Commercial parts. In applications for which no suitable corresponding AN, MS, or MIL type is in effect on date ofinvitation for bids, commercial parts may be used provided they conform to the requirements of this document.

3.5 Materials. Materials shall be as specified in this document and in MIL-E-4158. All materials shall be free of any defects and imperfections that may effect the serviceability and appearance of the finished product.

3.6 Design. The leak detector shall be designed to meet the requirements of this specification and MIL-E-4158. In areas of conflict between this specification and MIL-E-4158, the requirements of this specification shall govern. In areas where design criteria and performance have not been specified, the best commercial standard or practice shall be provided. The design shall minimize wasted space and give maximum facility of operation and maintenance. The leak detector shall weigh no more than 25 pounds and shall be operable by one man. The power supply shall be capable of operating the sensing unit a minimum of 4 hours without recharging. The leak detector shall be explosion-proof for operation in an environment defined by the National Electrical Code, NFPA 70, as class I, division 1, group D. The leak detector shall be safe for use in an explosive atmosphere at 8000 feet altitude and 125 degrees (deg.) Fahrenheit (F). The leak detector may be enclosed in one or two housings. In either case, the weight of the hand-held unit for leak searching shall not exceed 8 pounds. The dimensions of the housings shall not exceed 6 inches wide by 12 inches long by 10 inches deep. If two separate housings are utilized, the connecting cord shall be 8 feet in length with provisions for extending the working distance an additional 8 feet. The housings shall incorporate one common static electricity grounding point.

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A retractile (coiled) cable and spring clip shall be attached to that ground point for grounding to the aircraft. The cable retracted length shall be 12 inches. The extended length shall be 6 feet. The leak detector shall be furnished complete with all items and accessories required for proper operation.

3.6.1 Maintainability. Features providing the most expeditious and easy maintenance of the equipment consistent with performance, reliability, interchangeability, and cost, shall be used to the maximum extent practicable. The use of special tools not normally available commercially, and the need for special maintenance training shall be kept to a minimum.

3.7 Details of components.

3.7.1 Sensing unit. The sensing unit shall be a solid state device capable of detecting the following gasses at the specified leak rate:

Tracer gas	Leak rate
Helium	$1X10^{-3}$ cubic centimeter per second.
Argon	$4X10^{-3}$ cubic centimeter per second.
Nitrous oxide	$4X10^{-3}$ cubic centimeter per second.

3.7.1.1 Display and signal. Presence of the tracer gas shall be indicated on an electrical meter supplemented by a variable intensity light or an audible signal. If an audible signal is utilized it shall be mounted on or near the probe so that the probe tip and the light are visible simultaneously. The light or audio indication shall be visible in daylight or audible in earphone at 30 percent $\pm$5 percent full scale deflection. The specified leak rate of the tracer gas shall give a full scale deflection on the electrical meter. The electrical meter shall have sufficient displacement from zero to full scale to allow for easy determination of 25 percent, and 75 percent of the full scale reading. The display signal shall be a single displacement signal when the tracer gas is detected. Multiple displacement signal is not acceptable. Inherent electrical noise shall cause no more than 5 percent meter displacement. Zero drift shall be no more than 5 percent.

3.7.1.2 Weight distribution. If the sensing unit is to be hand-held during operation, the weight shall be distributed for easy, untiring, one-hand operation.

* 3.7.1.3 Response time. When utilizing the regular sampling probe (minimum length of one foot), the sensing unit full scale deflection response time to a $1X10^{-3}$ cubic centimeter per second helium leak shall not exceed one second.

3.7.1.4 Recovery time. When removed from a $1X10^{-3}$ cubic centimeter per second helium leak, the sensing unit recovery time (meter return to 10 percent of full scale) shall not exceed 3 seconds.

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3.7.1.5 Interference due to background gases. The sensing unit shall maintain sensitivity when operating in an atmosphere containing a minimum of 0.4 percent by volume concentration of turbine fuel in accordance with MIL-T-5624 grade JP-4 fuel vapor. The sensing unit shall maintain sensitivity when operating in an atmosphere containing a minimum of 5 percent by volume concentration of any of the tracer gases.

* 3.7.1.6 Automatic range capability. The sensing unit shall be capable of locating a leak of 1 cubic centimeter per second helium and a leak of $1X10^{-3}$ cubic centimeters per second helium when the leaks are 4 inches apart horizontally. there shall be evidence that two separate leaks have been located. This shall be accomplished in still air without utilizing range switches, zero adjustment, or other controls.

3.7.2 Probe. The probe shall be rigid or semi-rigid to allow for precise location and movement of the probe tip. Tubing or hose conveying the tracer gas from the probe to the sensing unit shall be flexible. The tubing or hose shall be sheated or of sufficient wall thickness to prevent crimping during search operations. The probe shall be no less than 12 inches and no more than 20 inches in length.

3.7.3 Power supply. The power supply shall be an internal power source capable of operating the leak detector for periods of 4 hours continuously. The batteries shall be scaled rechargeable type insensitive to orientation. Batteries shall be capable of a minimum of 500 recharging cycles from the voltage cut-off depth of discharge to fully charged condition. The power supply shall incorporate a voltage cut-off or other automatic device to prevent erroneous readings or indications due to low voltage. The power supply shall include an internal recharged capable of recharging the batteries from a 110-120 volts AC, 50-60 cycle outlet. The charger shall be capable of recharging the batteries from the voltage cut-off discharge to a fully charged condition in no more than 14 hours. Provisions shall be provided to prevent battery damage through over-charging. Charging will be accomplished in a non-hazardous environment. A detachable 3-wire charging cord shall be furnished. Male connection of charging cord shall conform to W-C-596, style D. The cord shall be at least 6 feet in length.

3.7.4 Carrying case. The carrying case shall weigh no more than 25 pounds. It shall be designed to provide storage for the leak detector and accessories and to provide protection during shipping.

3.7.5 Connecting cable (two housing type). If two separate housings are utilized, the power cable connecting the housings shall be three-conductor, extra-hard usage, flexible, type S, SO or ST cable with grounding conductor. The cable shall be attached to the housing so that a direct pull of 150 pounds will not produce a strain on the conductor connections or cause damage to the cable.

3.8 Electromagnetic interference. The leak detector shall conform to the requirements of MIL-STD-826 for class "T" equipment.

*3.9 Finish. The leak detector shall be impervious to turbine fuel in accordance with MIL-T-5624 grade JP-4 fuel. Manufacturer's standard color is acceptable.

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*3.10 Service conditions. The leak detector shall meet the requirements of MIL-E-4158. Operating temperatures will be 32° Fahrenheit (F) to 125°F.

*3.11 Part numbering of interchangeaable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part number and change thereto.

*3.12 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

*3.13 Workmanship. Workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3. All components, sub-assemblies, parts, and the final assembly shall comply with tests in section 4.

3.14 Hazardous location marking. The equipment shall be marked with the hazardous location for which it is approved or has been tested. If UL or factory mutual certified than the leak detector shall be marked in accordance with their procedures. If tested in accordance with MIL-STD-810, than the following marking shall be applied:

"Tested per MIl-STD-810, method 511.2, procedure II
explosion-proof in aviation turbine fuel atmospheres."

The marking shall meet the requirements of MIl-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Each as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of tests. The inspection and testing of the leak detector shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25° ±5° centigrade(C) and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure of temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.4 First article.

*4.4.1 Test samples. One leak detector shall be subjected to the first article inspection. In order to reduce the time required for first article testing, the contractor may elect to test more than one sample.

4.4.2 First article inspection. The first article inspection shall consist of all the tests described under 4.6. Test shall be performed in the order listed.

*4.4.3 First article test report. Upon completion of first article testing, a test report in accordance with MIL-STD-831 shall be prepared. The test report shall be authenticated by a responsible officer or the contractor and counter-signed by a Government witness. The test report shall explain how the test leaks were fabricated and calibrated. The test report shall contain a schematic diagram of the leak detector together with an explanation as to how the leak detector operates.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the individual tests.

4.5.1 Individual tests. Each leak detector shall be subjected to the following tests described under 4.6:

a. Examination of product (see 4.6.1).

b. Noise level and zero drift test (see 4.6.2).

c. Sensitivity test (see 4.6.3).

d. Response time test (see 4.6.4).

e. Recovery time test (see 4.6.5).

*4.6 Test methods. All tests involving any aspect of leak detection shall be performed with the probe tip no closer than 3/8 inch to the leak.

4.6.1 Examination of product. The leak detector shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, weight, size and merkings.

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4.6.2 Noise level and zero drift test. The leak detector shall be checked to determine that the inherent electrical noise causes no more than 5 percent meter displacement. During tests 4.6.3, 4.6.4 and 4.6.5 the leak detector shall be checked to determine that the zero drift is not more than 5 percent meter displacement. This check shall be for a minimum of 30 minutes.

4.6.3 Sensitivity test. The leak detector shall be exposed to a $1X10^{-3}$ cubic centimeter per second helium leak. A full scale deflection on the meter shall be recorded. This test shall be repeated two times (a total of three exposures). Any reading less than full scale shall be cause for rejection. The light and audio indicator shall be checked on each exposure for visibility or audibility and initiation at 30 percent +5 percent full scale meter deflection. This test (three exposure) shall be repeated for argon and nitrous oxide leaks of $4X10^{-3}$ cubic centimeter per second.

*4.6.4 Response time test. With the regular sampling probe attached, the leak detector full scale deflection response time to a $1X10^{-3}$ cubic centimeter per second helium leak shall not exceed one second.

4.6.5 Recovery time test. With the regular sampling probe attached, the leak detector shall be exposed to a $1X10^{-3}$ cubic centimeter per second helium leak. The probe shall be removed from the leak. Meter return to 0 + 10 percent full scale deflection shall not exceed 3 seconds.

4.6.6 Interference test (background gas). The leak detector shall be tested for compliance with 3.7.1.5.

4.6.7 Auotmatic range test. The leak detector shall be tested for compliance with 3.7.1.6.

*4.6.8 Power supply test. The leak detector shall be operated for 4 hours. During this 4 hour period the power supply shall be positioned 1 hour on each side and 1 hour in each end. The leak detector shall be exposed to a $1X10^{-3}$ cubic centimeter per second helium leak every 10 minutes. Sensitivity, response time and light or audio indication shall be as recorded in tests of 4.6.3, 4.6.4 and 4.6.5. Any evidence of malfunctioning shall be cause for rejection. The leak detector shall then be operated until the voltage cut-off or other sutomatic device is actuated. The recharger shall be activated in accordance with the manufacturer's instructions. The batteries shall be fully recharged in no more than 14 hours.

4.6.9 Electromagnetic interference test. The leak detector shall be tested in accordance with MIL-STD-826 for class T equipment.

4.6.10 Endurance tests. The leak detector shall be operated continuously for 30 hours. The 30 hour period shall consist of cycles of 1 hour "on" and 20 minutes "off." Four cycles would contribute 5 hours to the 30 hour total. Batteries shall be replaced after each 4 hours of operating. During each hour of operation the leak detector shall be exposed to a $1X10^{-3}$ cubic centimeter per second helium leak every 60 seconds. After completing the 30 hour endurance test, the leak detector shall be exposed to a $1X10^{-3}$ cubic centimeter per second helium leak. Sensitivity, response time, recovery time and light or audio indication shall be as recorded in tests of 4.6.3, 4.6.4 and 4.6.5.

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4.6.11 Connecting cable test. The leak detector shall be tested for compliance with 3.7.5. The 150 pound pull shall be applied for one minute. This test will be performed with the regular service cable and with the extension cable.

*4.6.12 Explosion-proof test. The leak detector shall be tested in accordance with procedure II, method 511.2 of MIL-STD-810. The test temperature shall be 125 deg. F. The test altitude shall be 8000 feet. The test facility shall be approved by the procuring activity. Either Underwriter's Laboratories Inc. certification as explosion-proof or factory mutual research approval as intrinsically safe for operation in a class I, division 1, group D environment is acceptable as meeting the explosion-proof testing requirement.

4.6.13 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The leak detectors shall be packaged method III in accordance with MIL-P-116, individually overboxed in unit containers conforming to grade W6C, PPP-B-636.

5.1.2 Level C. The leak detector shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the contractor's standard practice providing the latter meets the requirement of this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Leak detectors preserved and packaged as specified in 5.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size shall be limited to t he special requirements of Table II of PPP-B-636. Shipments that exceed 70 pounds shall be packed in overseas type containers conforming to PPP-B-601. Exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601 as applicable.

5.2.2 Level B. Leak detectors preserved and packaged as specified in 5.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Closure and strapping shall be in accordance with PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in interior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with uniform freight classification rules or regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection. Cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any other markings required in the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The leak detector covered by this specification is intended for locating fuel leaks in aircraft fuel cells and tanks using helium, argon, or nitrous oxide as a tracer media.

6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirement.

a. Title, number, and date of this specification.

b. Levels of preservation and packing (see section 5).

6.3 Suppression data. This specification includes the requirements of purchase description SANE 108.

6.4 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of one or more unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Key word listing.

Leak detector
Detector, leak
Fuel system leak detector
Helium leak detector
Electronic leak detector
Explosion-proof leak detector

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 4940-F546)

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